

SAN, Mali

WMO: 612770

Lat: 13.333N

Lon: 4.833W

Elev: 932

StdP: 14.21

Time Zone: 0.00 (E00)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	60.3	62.6	24.6	19.4	86.9	27.7	22.4	85.1	16.8	75.0	14.0	78.1	2.7	N/A	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	20.4	107.4	72.5	106.1	72.3	104.6	72.1	81.8	93.8	81.0	92.6	80.3	91.7	4.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	156.2	86.9	78.2	151.9	86.0	77.6	148.7	85.5	46.3	94.6	45.4	92.8	44.7	91.3	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
13.0	11.1	9.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.4	76.3	83.1	89.4	93.4	93.7	89.4	84.2	82.1	83.5	87.0	83.9	78.9
	DBStd	6.23	3.97	4.97	3.98	3.04	3.48	4.24	3.41	2.62	2.92	2.36	2.95	3.54
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12921	816	927	1222	1302	1354	1181	1060	996	1004	1147	1016	896
	CDD65	7446	351	507	757	852	889	731	595	531	554	682	566	431
	CDH74	101854	3789	6519	11574	14057	14487	10829	7355	5772	6423	9167	7165	4717
	CDH80	60568	1851	3843	7557	9786	10067	6726	3604	2354	2972	5162	4160	2486
Wind	WSAvg	3.1	3.8	4.2	3.3	3.4	3.9	3.9	2.9	2.3	1.9	1.9	2.6	3.6
Precipitation	PrecAvg	27.3	0.0	0.0	0.1	0.3	1.4	3.6	7.0	8.9	5.2	1.2	0.0	0.0
	PrecMax	35.2	1.0	0.2	1.3	2.1	3.8	7.0	12.9	18.6	10.9	7.0	1.0	0.6
	PrecMin	15.7	0.0	0.0	0.0	0.0	0.1	1.0	2.7	4.8	1.0	0.0	0.0	0.0
	PrecStd	4.9	0.2	0.0	0.2	0.4	0.9	1.6	2.4	3.0	2.4	1.3	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.0	105.3	107.7	109.4	108.9	105.9	100.2	94.9	97.2	101.9	101.6	99.0
		MCWB	63.8	66.7	68.1	71.7	74.2	77.1	78.6	79.7	80.4	72.1	68.1	66.6
	2%	DB	94.1	102.3	105.8	107.6	107.3	103.9	97.5	93.0	94.8	100.3	99.8	96.7
		MCWB	62.2	66.7	68.5	71.6	74.6	77.4	79.1	79.7	80.6	73.5	68.0	65.2
	5%	DB	91.8	99.8	104.2	106.1	106.0	101.8	94.9	91.3	93.2	98.7	98.4	94.7
		MCWB	61.1	65.7	68.1	71.3	74.7	77.7	79.0	79.3	80.1	74.9	68.0	64.2
	10%	DB	89.5	97.0	102.4	104.2	104.2	99.5	92.7	89.6	91.6	96.9	96.6	92.2
		MCWB	60.3	64.4	67.2	71.1	75.3	77.6	78.9	79.6	79.6	75.6	67.8	63.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.0	72.8	76.9	80.1	80.9	81.6	81.9	81.9	82.7	82.8	77.0	70.4
		MCDB	88.0	95.1	102.4	100.4	98.9	96.8	94.1	90.9	93.6	96.6	93.1	91.9
	2%	WB	64.7	69.8	74.4	78.1	79.9	80.6	80.8	81.7	81.2	75.1	67.8	66.2
		MCDB	89.2	92.9	96.6	98.1	98.0	96.0	92.2	89.3	92.1	94.2	89.3	90.3
	5%	WB	62.8	67.1	73.0	77.0	79.1	79.9	80.0	80.0	80.9	80.0	73.4	66.2
		MCDB	87.1	95.4	93.8	97.6	97.2	94.8	90.7	88.4	90.7	92.0	87.3	88.7
	10%	WB	61.3	65.4	71.3	75.9	78.3	79.2	79.3	79.4	80.2	78.9	71.7	64.7
		MCDB	86.2	94.6	93.1	96.4	96.5	93.7	89.5	87.4	89.3	90.3	87.1	86.7
Mean Daily Temperature Range	5% DB	MDBR	26.9	26.7	25.7	22.1	20.4	18.3	15.6	14.1	15.9	20.7	26.8	27.4
		MCDBR	29.8	29.2	27.3	24.3	22.2	20.2	18.2	16.3	18.0	23.2	28.4	30.0
		MCWBR	9.1	9.4	9.8	8.8	7.1	5.3	5.5	5.5	6.5	6.9	6.9	7.8
	5% WB	MCDBR	28.4	26.9	22.1	20.3	19.8	18.6	16.5	15.1	17.1	19.6	23.0	28.9
		MCWBR	10.3	10.7	9.9	7.4	6.3	6.2	6.0	6.3	7.0	7.4	7.8	8.2
Clear-Sky Solar Irradiance	taub		0.524	0.567	0.698	0.716	0.707	0.683	0.608	0.566	0.559	0.584	0.532	0.505
	taud		1.863	1.752	1.519	1.564	1.582	1.661	1.839	1.940	1.943	1.855	1.866	1.909
	Ebn at Noon		245	241	214	210	208	210	228	240	243	233	241	246
	Edh at Noon		63	73	95	90	87	79	67	61	61	65	62	59
All-Sky Solar Radiation	RadAvg		1748	1920	2032	2049	1992	1874	1814	1761	1873	1890	1794	1708
	RadStd		99	102	106	108	72	54	68	91	93	48	50	66

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page